SMART GRIDS PROJECT

Simulation-Based Spatial Optimisation of Solar PV for Loss Minimisation in Smart Grids Distribution Systems.

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Devanshu Shrivastava

Prof Amin Mahmoud
Smart Grids
Rutgers University, New Brunswick

Presentation Outline

Problem Statement and Motivation

5–13%

Power lost as heat
in distribution networks

70%

Of total grid losses
occur in distribution

↑ Solar

PV penetration rapidly
increasing worldwide

The Core Challenge

Traditional distribution networks **were designed for unidirectional power flow** — from substation to consumers. The rapid integration of **distributed solar PV** can introduce reverse power flows, voltage fluctuations, and increased losses if panels are installed **without spatial optimisation.**

Research Question: Where should Solar PV be placed, and at what capacity, to minimise active power losses?

Network Model: IEEE 33-Bus Distribution Network

33 Buses

Nodes where loads are connected

32 Lines

Branches carrying power between buses

12.66 kV

Base system voltage

3.715 MW

Total system load

Network Topology

- **Main Feeder:** Bus 1 → 2 → 3 → … → 18 (backbone of the network)
- **Lateral 1:** Branches from Bus 2 → 19 → 20 → 21 → 22
- **Lateral 2:** Branches from Bus 3 → 23 → 24 → 25
- **Lateral 3:** Branches from Bus 6 → 26 → 27 → … → 33 (heaviest loaded)

Bus 1 (Substation) is the Slack Bus — voltage fixed at 1.0 pu. All other buses receive power from here.

Network Model: IEEE 33-Bus Distribution Network

3.715 MW

Total System Active Load

202 kV

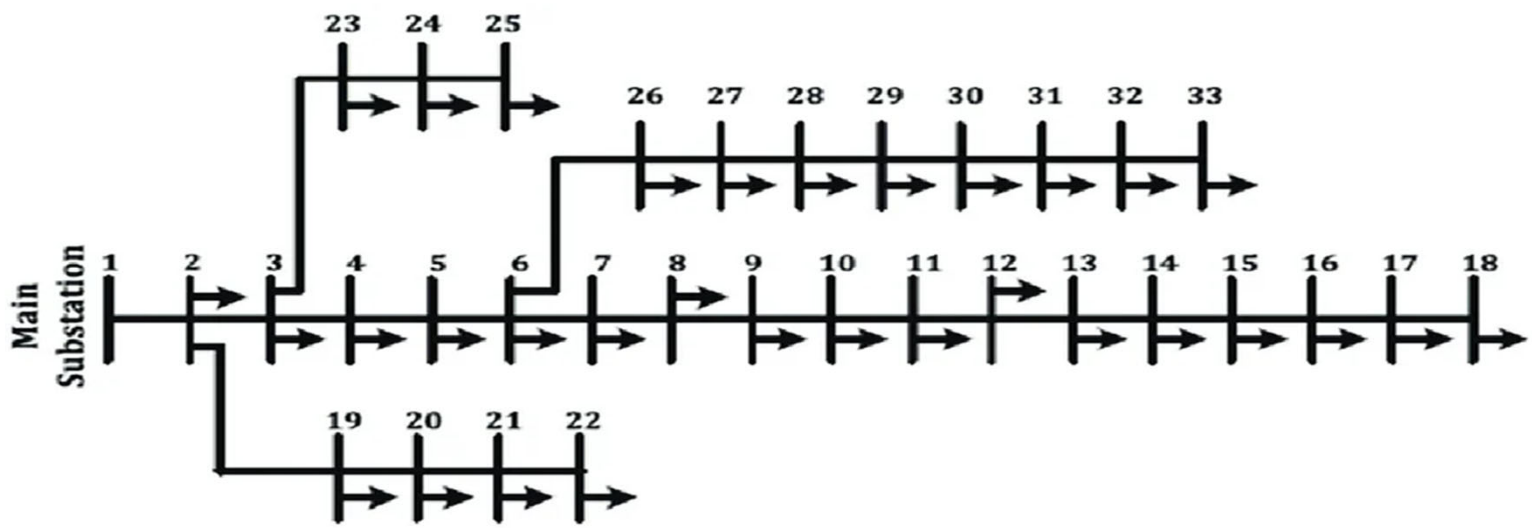
Baseline Active Power Loss

12.66 kV

Base system voltage

2.300 MVAR

Total System Reactive load



Methodology

1. **Network Setup**
   IEEE 33-bus line data loaded — resistance, reactance, and load at each bus defined as matrices in MATLAB

2. **Power Flow (DistFlow)**
   Backward/Forward Sweep using Baran & Wu DistFlow equations solves voltages at all 33 buses and computes baseline losses

3. **PV Modelling**
   Solar PV modelled as negative load (power injection) at candidate bus. Reactive power injection set to zero (unity power factor)

4. **Spatial Sweep**
   PV unit (fixed 1 MW) placed at each bus 2–33 sequentially. Power flow re-run each time. Losses recorded per location

5. **PSO Optimisation**
   Swarm of 30 particles simultaneously optimises bus location (2–33) and PV size (100–3000 kW) over 100 iterations

Particle Swarm Optimisation (PSO)

Biological Inspiration

Inspired by the collective foraging behaviour of bird flocks. Each particle (bird) remembers its personal best position and is attracted toward the global best found by the entire swarm.

PSO Parameters Used

Particles	30
Iterations	100
Bus search space	2 – 33
PV size space	100 – 3000 kW

Velocity Update: $v = w \cdot v + c_1 \cdot r_1 \cdot (\text{personal\_best} - \text{position}) + c_2 \cdot r_2 \cdot (\text{global\_best} - \text{position})$

Inertia ($w \cdot v$)

Keeps particle moving in current direction — encourages exploration of new areas

Personal Pull (c_1)

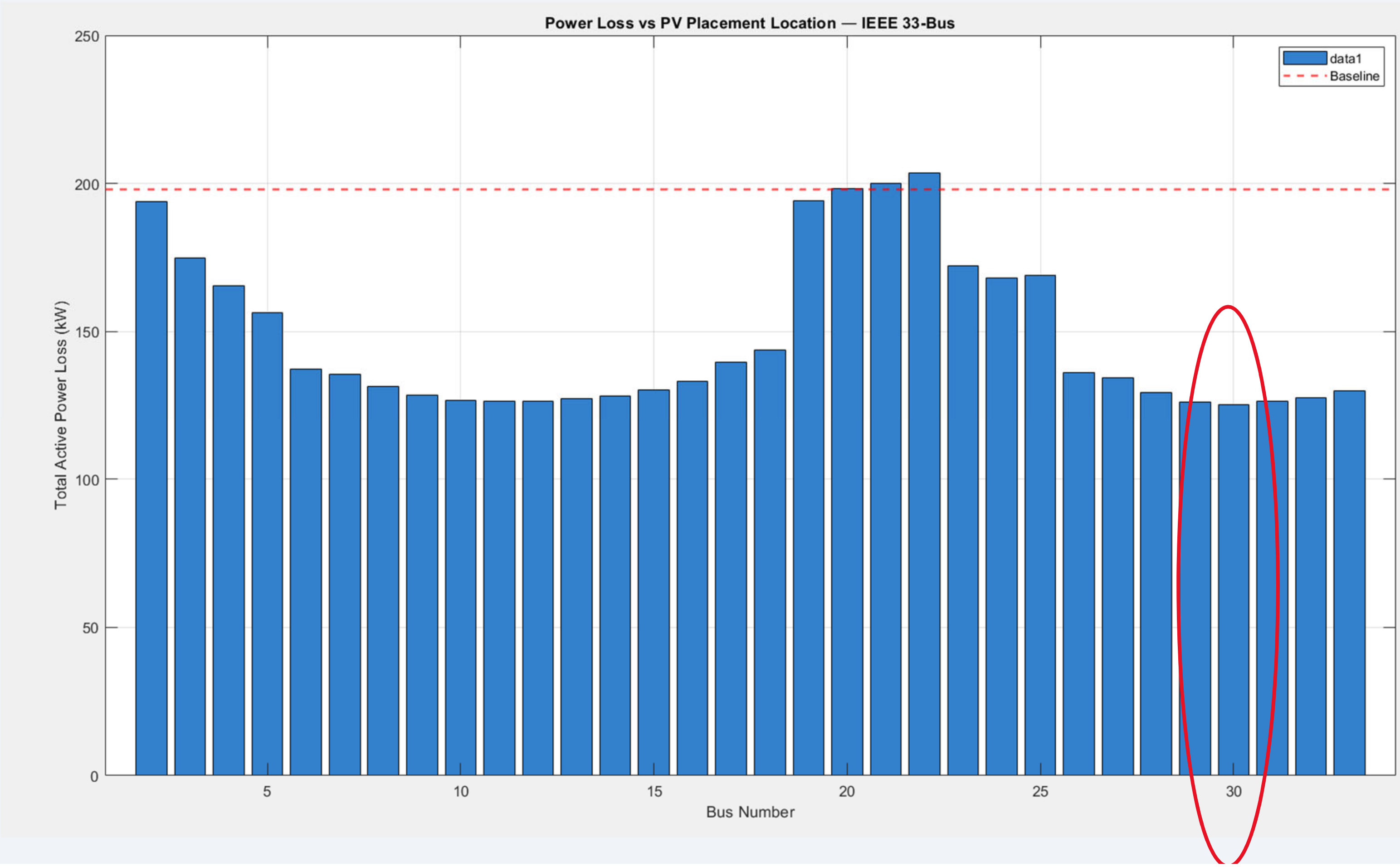
Attracts particle back to its own best known position — memory of past success

Social Pull (c_2)

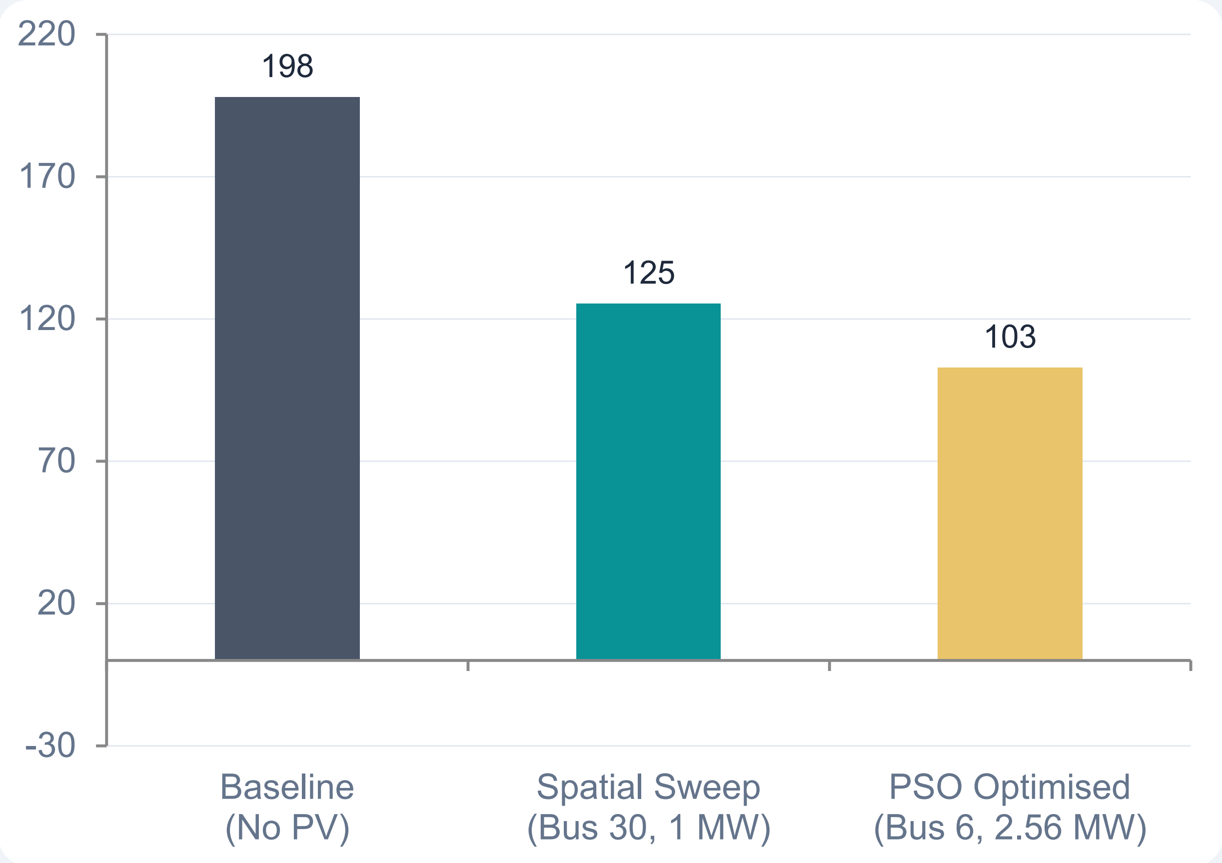
Attracts particle toward swarm's global best — collective intelligence

Results — Power Loss Comparison

Spatial Sweep: Minimum Active Power Loss: Bus 30 = 125.3 kW



Results — Power Loss Comparison

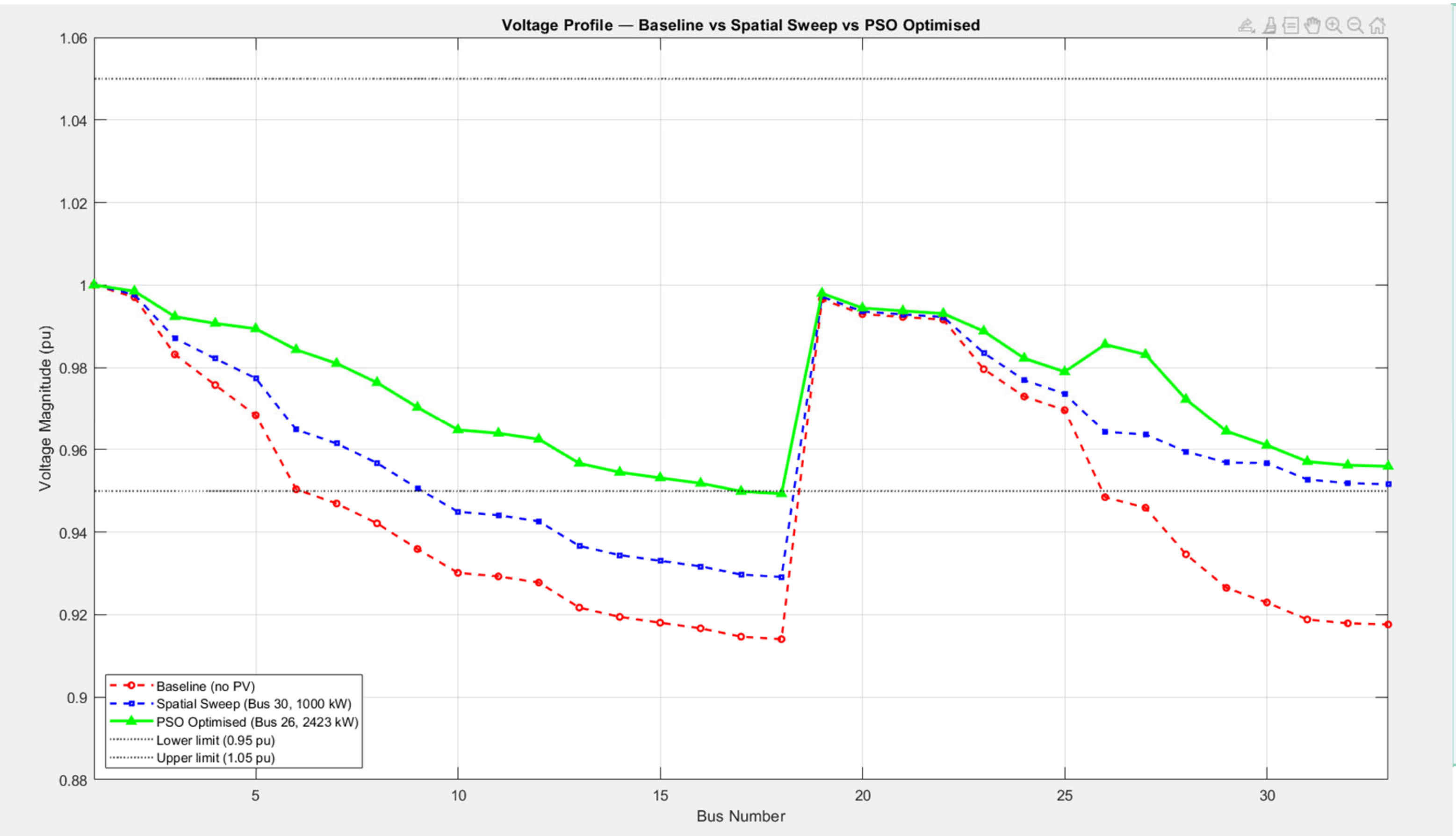


36.7%
Loss reduction
Spatial Sweep

48.0%
Loss reduction
PSO Optimised

0.94 kW
PSO std deviation
(6 independent runs)

Results — Voltage Profile Analysis



Key Observations

- 🔴 Baseline dips to 0.917 pu — violates IEEE 0.95 limit at 12 buses
- 🔵 Spatial sweep improves main feeder but lateral violations remain
- 🟡 PSO profile stays above 0.95 pu across most of the network
- 📍 Bus 19 spike visible — close to substation on Lateral 1
- ✅ PSO reduces voltage violations from 14 buses to near zero

PSO Sensitivity Analysis — 6 Independent Runs

Run	Optimal Bus	PV Size (kW)	Loss (kW)	Reduction
1	Bus 26	2422.76	104.67	**47.12%**
2	Bus 6	2561.54	102.84	**48.04%**
3	Bus 6	2561.54	102.84	**48.04%**
4	Bus 26	2422.76	104.67	**47.12%**
5	Bus 6	2561.54	102.84	**48.04%**
6	Bus 6	2561.54	102.84	**48.04%**

103.45 kW
Mean Loss

0.94 kW
Std Deviation

102.84 kW
Best Loss

47.73%
Mean Reduction

Low standard deviation (0.94 kW) confirms PSO reliably converges despite random initialisation.

Conclusions & Key Findings

47.1% Loss Reduction

PSO optimised placement at Bus 6 (2.56 MW) reduces active power losses from 197.9 kW to 102.8 kW

Voltage Violations Resolved

Baseline had 14 buses below 0.95 pu limit. PSO placement brings network close to compliance throughout

Location Matters More Than Size

Spatial placement at feeder branch entry points (Bus 6, 26) outperforms deep-network placement regardless of capacity

PSO Robustness Confirmed

6 independent runs yield std deviation of only 0.94 kW — algorithm reliably converges to near-optimal solutions

Future Work: Multi-unit PV placement, battery storage integration, dynamic load profiles

THANK YOU! QUESTIONS?